AgriCrop: Geospatial Plant Disease &Soil Moisture Intelligence Network

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Project Title:

AgriCrop: Geospatial Plant Disease & Soil Moisture Intelligence Network

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Project Overview

Problem Statement: Smallholder farmers face crop losses because plant diseases and soil moisture issues are not identified early.

Objective: Build an AI-powered system for plant disease detection and soil moisture prediction using machine learning and deep learning.

Technologies Used: Python, TensorFlow, Flask, Scikit-learn, HTML, CSS, JavaScript, MongoDB, Leaflet.js.

Dataset Details

Plant Disease Dataset (PlantVillage):
https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset

Alternative: https://www.kaggle.com/datasets/mohitsingh1804/plantvillage

Soil Dataset: https://www.kaggle.com/datasets/r3trovision/soil-moisture-temp-and-nutritions

Project Workflow

1. Data Collection
2. Data Preprocessing
3. Plant Disease Image Classification using MobileNetV2
4. Soil Moisture Prediction using Machine Learning
5. Model Training and Evaluation
6. Generate classes.json
7. Flask Integration
8. Image Upload and Prediction
9. Display Results and Recommendations
10. Future Deployment with Geospatial Dashboard

Model / Algorithm Used

MobileNetV2 CNN for crop disease identification from leaf images and Random Forest Regression for soil moisture prediction. MobileNetV2 was selected because it is lightweight and provides high accuracy for image classification tasks.

Implementation

Implemented a Flask-based web application that allows users to upload crop leaf images and receive disease predictions. Trained and saved deep learning and machine learning models for deployment.

Results

Disease Model Validation Accuracy: 93.16%
Soil Moisture Prediction Accuracy: 78.81%
Successfully predicts plant diseases and estimates soil moisture levels.

Challenges Faced

• Installing TensorFlow and resolving environment issues.
• Managing large datasets and long training times.
• Handling file path and image upload errors.
• Model integration with Flask.
• Organizing project structure and dependencies.

Conclusion

Successfully developed an AI-based agriculture assistance system capable of predicting plant diseases and soil moisture levels. Future improvements include real-time geospatial mapping, recommendation systems, and cloud deployment.

References

Kaggle Datasets, TensorFlow Documentation, Flask Documentation, Scikit-learn Documentation, and online tutorials/articles.

Submission Files

1. Project Report (PDF)
2. Source Code (Github Repository Link)

https://github.com/himabindu0322/AgriCrop-Geospatial-Plant-Disease-Soil-Moisture-Intelligence-Network